

Six legs: scale only what has been verified

M23 • 01 Oct 2028 - 01 Nov 2028 • 75 hours

The completed robot

M23 aims to produce **a six-legged robot v1.0 with 18 degrees of freedom, slow walking and active body levelling**: three actuators per leg. It must stand independently for at least 120 seconds, perform wave and ripple gaits under declared conditions, and diagnose loss of data, contacts and individual components. The central question is what changes when moving from four legs to six in mechanics, power, communication timing and support control.

18 DOF refers to the number of actuated joint coordinates, not the number of independent body motions available on the ground. Additional legs provide more contact-allocation options, but increase mass, energy consumption, wiring and possible faults. Simply appending six IDs to an array does not validate previous calculations. Analysing the architectural change is a major practical task.

Entry check before scaling

Open the M22 package: independent quadruped standing, slow gait, functioning independent stop, calibrations, current and bus budgets, and analysis of significant failures. Record the entry decision PASS/HOLD in writing. If E-stop, power, actuator identification or the control cycle remain unverified, physical expansion stays on HOLD. In that case, complete M23 calculations and simulation tasks while resolving hardware blockers.

The six additional actuators and parts for two legs are planned only after this check, as in the original plan. Purchasing is not automatic: first verify compatibility, calculate the budget and obtain the project owner's decision. Changing actuator family for the last two legs requires new calibrations, a delay model and verification of joint operation.

The operating scope of the result

The $\pm 10\text{-}15^\circ$ range in the overall plan is a design target for body levelling and slope tests, admissible only where the particular assembly's geometry, stability, traction and loads permit. Do not adopt it as an operating range before calculation. Start with a smaller verified angle and expand only when criteria are met. State surface angle and desired body angle separately: they are different quantities.

The robotic hand develops along its own OPT workstream after M17; M23 leg verification does not replace its limits or results. Shared logging and diagnostics benefit both systems. The final goal is a walking robot and hand by programme completion, so pass six-legged-version limitations and unfinished hand items to M24 and the separate G4 without adding hidden hours.

Work blocks, measurements and scope management

The monthly plan: 35 hours of analysis + 40 hours of practice

1-20: 10 hours recalculating mechanics, power and communication; 10 hours assembling, calibrating and testing 18 devices. **21-40:** 10 hours on contact sequences and admissible transitions; 10 hours implementing wave and ripple in simulation and on the prepared test area.

41-60: 10 hours on surface, contact and uncertainty estimation; 10 hours on active levelling, bounded terrain and operation without a camera. **61-75:** 5 hours analysing reliability and energy; 10 hours conducting a test series, maintenance and preparing v1.0.

Organise the first three analysis blocks as five 2-hour sessions each; the last is 2 + 2 + 1. Practice comprises four Sunday blocks of 10 study hours. Alternate movement, cooling, inspection, resetting initial conditions and analysis. Charging time alone is not study time; use that interval for calculations and data processing.

One experiment, one question

For every run, record gait type, phase order, cycle duration, swing height, target body tilt, surface, load, battery condition and temperatures. Begin by comparing two configurations differing in one parameter. Simultaneously changing gait, surface and controller gain prevents you from identifying why error changed.

The first-page dates repeat the overall plan's calendar window. Shared boundary dates do not double hours; holidays and integration G blocks retain their place in the overall schedule.

Work-hour numbers specify task order and do not require extra evening work. Print parts and arrange components in advance after entry-check authorisation, allowing for actual delivery time.

Reducing scope while preserving the objective

If behind schedule, postpone tripod first, then sideways motion on difficult terrain, then optional depth-camera comparisons and additional surfaces. Retain wave and ripple, standing, independent stopping, admissible-slope assessment, operation when the camera is unavailable and transparent repeated-test results. Without hardware verification of wave and ripple, call the result a candidate, not a completed six-legged robot v1.0.

Do not consume the hand reserve or G4's 35 hours on minor gait refinements without replanning the programme. The main deliverable is a justified operating envelope. Python is useful for support geometry and reports, C++ for the existing control loop. An additional log analyser may be written in Rust; a third implementation is optional.

The six-legged frame and the 18-actuator budget

What to study and recalculate

Examine the arrangement of six hip assemblies, fastener access, harness routing and centre-of-mass changes. Check overlap between adjacent leg workspaces, load transmission from the middle pair to the frame and whether one actuator can be removed without dismantling the body. New legs use the existing naming conventions and units but have their own calibrations. Estimate stiffness for the selected design load case and compare it with admissible foot-position change.

Build the power budget for actual load combinations: standing, initiating swing, a low body posture and stopping. Do not assign each leg exactly one sixth of the weight in every phase. List specification limits and cooling conditions for branches, connectors, BMS and DC-DC converters. Summed rated peaks represent only one conservative scenario; separately measure simultaneous currents and their duration.

Task 1. Design review: 4 → 6

Inputs are the M22 candidate, six-legged-frame CAD and component specifications. 1) List changes to parts, masses, axes, harnesses and power distribution. 2) Check fasteners, clearances and service access for every new hip frame. 3) Calculate the new configuration's centre of mass and predict support reactions. 4) After unpowered assembly, measure mass and verify geometry using reference dimensions.

Save `scaleup_design_review.md` and an updated `assembly_manifest`. Check that every change traces to a CAD revision and that software renaming does not conceal workspace collisions. Boundary case: a middle foot collides with a front link during turning. Correct the trajectory or geometry and save the forbidden region as a test before permitting physical motion.

Task 2. Power and communication after expansion

Calculate the 18-actuator communication cycle for your protocol and telemetry content, then compare with M22 measurements for 12 devices. With the robot supported, add new branches in stages, checking IDs, polarity, signs, zeros and idle current. Measure simultaneous communication at the actual frequency, computer load and enabled diagnostics. Record supply and branch currents separately where instruments permit.

Save `power_bus_18.csv` and plots of margin to the communication deadline. Acceptance: new devices create no duplicate IDs, and voltage sag and delays remain within justified operating ranges. Boundary case: two branches work separately but overload a shared DC-DC converter together. Revisit calculation and measurement of the common circuit; reducing gait speed is not a universal solution to an electrical problem.

Wave, ripple and admissible contact transitions

Gait model and transition conditions

Describe each leg with the state machine STANCE → LIFT → SWING → TOUCHDOWN → STANCE. In STANCE, the foot is an assumed support; in SWING it moves; TOUCHDOWN requires contact confirmation. Transitions depend on more than time: check target reachability, state freshness and remaining supports. Phase ϕ from 0 to 1 identifies position in the cycle, while duty factor is the fraction of the cycle assigned to stance.

For slow motion, calculate the convex polygon of confirmed supports and the centre-of-mass projection along gravity onto their plane. Support margin is the signed minimum distance from this projection to the polygon boundaries. On a horizontal floor, this is the ordinary vertical projection; retain the projection direction on a slope. Positive geometric margin alone does not check friction, actuator torque or dynamics.

Task 3. A wave table and verifiable support margin

Inputs are the six-legged model, level floor, calibrated axes and initial standing pose. 1) Specify sequential single-leg swing and a phase table for all six legs. 2) Before lift-off, recalculate remaining supports and body-position feasibility. 3) Visualise contacts, polygon and margin in metres. 4) Run a complete cycle, startup and stopping, first in simulation and then with a small permitted step on the test area.

Save `gait_wave.yaml`, the phase diagram and a log of minimum margin. Check that no more than one leg swings at a time, feasible support exists before lift-off and commands cannot skip touchdown. Boundary case: the foot has not touched down but the timer requests the next lift. Block the transition and emit an event; the state machine must not continue without contact confirmation.

Task 4. Ripple, turning and sideways motion

Construct ripple with more overlap between swings, an explicit phase table and allowed combinations. Do not rely on the gait name alone: ripple implementations use different conventions. For every combination, check reachability, collisions and support. Then add turning about body z and sideways translation separately; positive directions follow the frozen coordinate frame.

Save `gait_ripple.yaml`, allowed-transition tests and three short scenarios: forwards, turning and sideways. Check that an identical command has the same meaning for right and left legs and that mode changes pass through an admissible state. Boundary case: switching wave → ripple during swing. Permit it only after phase alignment or through a stop. If the assumed contact is physically impossible, the test fails even when motion appears smooth.

Six contacts, the surface plane and uncertainty

What to study

Review transforming foot coordinates from leg frames to a common body or world frame. For an approximately planar patch, estimate $z = a \cdot x + b \cdot y + c$ from confirmed supports: matrix A contains rows $[x, y, 1]$, vector z contains foot heights in metres, and the solution minimises squared residuals. The normal is proportional to $(-a, -b, 1)$. Use weights for measurements with different reliability, explaining their origin.

A plane model cannot represent arbitrary stairs or separate rocks without examining residuals. Three nearly collinear points provide poor geometry even with a small computed error. Six coordinates are not six independent measurements if all derive from one erroneous body orientation. Contact, a stationary foot and estimator confidence are three different indicators.

Task 5. Estimate a plane from six supports

Generate synthetic data for a horizontal plane and two inclined planes with known parameters, then add noise, one outlier and contact loss. 1) Transform every foot into one frame. 2) Select confirmed supports. 3) Solve by least squares and check rank and conditioning. 4) Calculate residuals and a validity flag. 5) Compare with an independent reference measurement of the prepared area.

Save `terrain_estimator_tests` and `plot_plane_error`. Check that ideal data recover the specified plane within numerical precision; with noise, compare error to the declared tolerance, not zero. Boundary cases are two supports, nearly collinear contacts and one foot on a protrusion. Return invalidity or a limited estimate in these cases; an arbitrary normal without a flag is unacceptable.

Task 6. Depth-based foothold selection and camera failure

Use a saved scene depth map and camera calibration from earlier modules. Construct a small region of admissible footholds: reachable kinematics, clearance, known surface, height uncertainty and image age. Compare selection against a method using only confirmed contacts. With the camera disabled, the robot remains in a previously tested mode or stops; an IMU alone does not authorise movement over unknown terrain.

Save `foothold_candidates.json` and two identical scenarios with and without the camera. Check that unknown depth does not become zero height, a stale image is not treated as fresh, and the selected foothold passes reachability checks. Boundary case: a shiny or dark surface creates a hole in the depth map. Show uncertainty and explain rejection of the placement while retaining the classical fallback mode.

Active levelling and the terrain operating boundary

Relating body tilt to leg configuration

Levelling maintains a selected body orientation relative to gravity by changing leg positions within reachability and load limits. The IMU estimates orientation, contact estimation describes support, and kinematics converts desired body position into joint commands. Zero roll and pitch do not mean the feet lie at the same height; on a slope, leg configurations may differ.

Map the constraints for each pose: joint travel, proximity to singularity, loaded torque, body clearance, traction, contact quality and support margin. Estimate quasi-static joint torque using $J(q)^T \cdot f$, where J is the foot Jacobian, f is force in newtons and the result is in N·m. Force allocation must satisfy equilibrium and contact constraints; equal forces on all six legs are only one special case.

Task 7. Verify 120 seconds of standing

Start on a level surface in a previously verified configuration. 1) Run a short current and contact check. 2) Stand independently for at least 120 seconds without the backup support carrying weight. 3) Record roll and pitch errors, data freshness, minimum support margin and temperatures. 4) Repeat after cooling and restoring initial conditions, explicitly identifying every repeat.

Save stand_120s with raw log and video. The criterion is a complete continuous interval within declared limits, without body support or unaccounted measurement losses. This verifies a particular standing condition, not machine lifetime. Boundary case: tilt error decreases by saturating one joint. Body-orientation error is small, but a constraint is violated, so the trial fails.

Task 8. Map slopes and uneven terrain

Before the physical trial, calculate admissible surface angles and commanded body orientations, then test them in simulation with mass, friction and delay variations. On an adjustable rig with suitable backup support, increase slope in steps from a verified value. For each angle, save achieved body roll and pitch, currents, joint poses, contacts and stopping reason. Introduce unevenness as a separate factor with measured height.

Save operating_envelope.csv: tested, untested and forbidden regions. Mark the $\pm 10\text{-}15^\circ$ design target as achieved only from actual tests of the specific configuration. If the limit is smaller, identify the limiting mechanism. Boundary case: traction is lost while torque remains admissible. Reducing controller error does not solve a contact problem. Return to level ground and investigate foot-surface interaction.

When expanding the M20 force allocator, update contact-set size, constraints and solver timing measurements. In HIL, test infeasible constraints and solver-deadline overruns: retain the previously tested classical fallback. The M21 policy is outside the mandatory physical result; transfer from four legs to six requires separate verification of observations, actions and applicability.

Reliability, energy and maintenance

A fault tree with a clear top event

Choose the top event: the robot loses admissible support or fails to stop on request. Divide causes into power, bus, actuator, mechanical connection, state estimation and control logic. AND denotes a combination of required events; OR denotes alternative sufficient paths. Do not assign branch probabilities without data; a qualitative tree already helps select tests. A common power source or configuration error links several legs, so do not silently assume independent failures.

With only five functioning legs, the system must first recognise the condition and restrict behaviour. Continued walking depends on geometry, load and how the sixth leg failed. A freely dangling leg, a leg jammed in the air and a leg jammed while supporting load require different responses. The basic fallback to verify is stopping or lowering according to the design procedure. Five-legged walking requires a separate justification.

Task 9. Component or leg failures and recovery

From the tree, select telemetry loss, a stopped node and incorrect contact. First inject them in simulation or HIL, recording the stimulus time, detection and resulting state. Then reproduce only permissible rig variants with the robot supported. Check that a recovered node cannot receive an old command without state verification and an explicit resumption procedure.

Save `fault_tree` and `fault_runs` with a timeline. Check that every stimulus has an expected outcome and measured detection time; behaviour must fit the remaining contacts and limits. Boundary case: a common bus failure produces six identical warnings instead of recognising a shared cause. Add aggregate status while retaining primary errors. Do not unplug a loaded connector by hand to demonstrate a fault.

Task 10. Energy and maintainability

On a marked route, measure voltage $U(t)$, supply current $I(t)$ and distance d . Energy $E = \int U \cdot I \, dt$ is in joules; $E/3600$ is Wh, and E/d is J/m for a valid non-zero distance. Compare wave and ripple at comparable speed, surface, mass and temperature. For a stationary robot, report power and energy over a selected interval; do not divide by a near-zero distance.

Save `energy_comparison.csv` with integration boundaries, measurement gaps and repeats. Then, with the system de-energised, replace an accessible service assembly according to its instructions, measure the time, and repeat inspection and calibration. Save `maintenance_runbook` and a reasonable spare-parts list based on actual failures. Boundary case: the mechanics are restored but zero is lost. Restart is prohibited until the joint record is updated.

For thermal balance, tabulate every actuator's temperatures on the same route: initial, maximum observed, rise and load duration. Identify a consistently overloaded leg and check zero, friction and support allocation. Change only one admissible configuration or phase parameter and repeat; reduced heating in one leg must not conceal increased heating elsewhere or compromised support.

The test series and six required boundary cases

A repeat plan with a defined trial unit

Plan 40 attempts as a working target within the original 30-50+ guideline. Example allocation: 10 wave trials on level ground, 10 ripple trials on the same ground, 10 levelling trials on a verified slope, and 10 transitions into restricted operation without a camera. Each attempt includes restoring the starting pose, checking conditions, executing the specified scenario and recording the outcome. Sensor samples and steps within one traversal are not independent attempts.

If equipment availability or cooling duration prevents this plan, justify a smaller sample and narrower conclusion in advance. Do not replace unavailable difficult trials with many identical easy ones while claiming the original coverage. Mark untested conditions explicitly. For comparisons, fix the order or alternate modes so battery discharge is not mistaken for a gait advantage.

Control scenarios

T1. Camera loss during an admissible short route: enter a verified fallback or stop, without selecting new unknown footholds. **T2.** One false contact during replay: the estimator accounts for inconsistency and does not automatically expand support. **T3.** Nearly collinear surface points: return poor-conditioning or invalidity status; do not present an arbitrary slope as a reliable estimate.

T4. Stale state from one leg: do not authorise the next phase using an old pose. **T5.** Wave/ripple switching in an inconvenient phase: defer the transition or pass through a stop, retaining an admissible contact order. **T6.** A stopped node or bus loss in HIL: detection, restricted behaviour and recovery proceed without unintended resumption. A physical variant requires impact analysis and support verification first.

What the results can establish

Report N, successes, failures and unusable records separately by condition. Calculate a confidence interval for success rate and explicitly state the assumption of independent trials. Even 40 successful tests do not prove absolute reliability or determine lifetime in hours. Temperature drift and a shared defect in one assembly make trials dependent; discuss this limitation instead of artificially increasing N.

The summary must include at least one negative scenario with the expected restriction of behaviour and one real failure if one occurred. Analyse causes, add a regression test and repeat the relevant test class. A version change during the series separates data into different configurations; do not combine them into one v1.0 estimate without explanation.

The first 40 hours: expansion and gaits

Hours 1-10: decisions and calculations

1-2: check the M22 package and record PASS/HOLD for physical expansion. 3-4: compare four- and six-legged CAD and map loads, service access and conflicts. 5-6: recalculate mass, centre of mass and power scenarios. 7-8: assess BMS, DC-DC converters, protective branches and communication margin for 18 actuators. 9-10: specify assembly, calibration and E-stop re-verification sequences.

Put unresolved specification parameters in a question table. For each, identify its decision impact: blocks power-up, limits the tested mode or concerns future expansion. Separately assess the effects of external covers or styling and unknown power-connection ratings. By the end of hour ten, have a concrete task 1 work list and expected task 2 measurements.

Hours 11-20: expansion on the rig

11-14: assemble new hip frames and two legs; measure mass, add labels and check clearances. 15-17: staged connection, identification, signs and calibration of new joints. 18-19: communication load and power measurements with 18 devices. 20: repeat independent stopping, inspect and freeze the revision. If the entry decision is HOLD, use these hours to repair the quadruped and test the six-legged model without physical expansion.

Post-assembly verification must include the original four legs: new harnesses or added mass can impair previously working components. Repeating known M22 scenarios checks preservation of earlier functions, but does not verify an untested six-legged gait. Update numerical limits only with supporting evidence.

Hours 21-40: contacts and movement

21-22: phase state machine and duty factor. 23-24: support polygon, centre-of-mass projection and permissible-lift criteria. 25-26: wave table and touchdown that does not occur when the timer expects it. 27-28: ripple and allowed overlapping swings. 29-30: turning, sideways motion and gait changes. For every 2-hour session, prepare one test or drawing needed in practice.

31-34: task 3 in simulation, support checks and a short hardware wave traversal after rig testing. 35-37: task 4, ripple and transition through an aligned phase. 38-39: separate turning and sideways scenarios within the verified region. 40: analyse worst support margin and correct one transition fault. Treat tripod as a possible extension; do not add it at the expense of unfinished touchdown control.

Hours 41-75: terrain and evidence

Hours 41-50: four important distinctions

41-42: foot coordinate frames, gravity and the support plane. 43-44: least squares, residuals, rank and nearly collinear contacts. 45-46: contact confidence, slip and body observability. 47-48: depth uncertainty, image age and selection of a reachable foothold. 49-50: load and kinematic-constraint maps on a slope, desired body orientation and conditions for restricting motion.

Explain four distinctions in writing: touching versus stationary support; the plane beneath the feet versus the horizontal; body tilt versus surface slope; and no camera versus no sensors at all. If these concepts are confused, postpone new-terrain trials and correct notation and tests. Terminology must agree across code, plots and protocol.

Hours 51-60: the surface experiment

51-53: task 5 on synthetic data and the measured area, including an outlier and contact loss. 54-55: task 6, the saved depth map and the same scenario without a camera. 56-57: preliminary and qualifying 120-second standing for task 7, with analysis. 58-60: stepped slope increases and small measured terrain variations for task 8, only within the admissible region. Save every attempt, but count it in the confirmation series only if it follows the predefined protocol.

A failed slope trial still yields a useful result: the operating boundary and its cause. Do not raise current limits to reach a desired plot value without recalculating the complete power circuit and thermal behaviour. Do not change foot size midway through a series without a new design revision either: traction and the contact model then change.

Hours 61-75: reliability and release

61-62: fault tree, five functioning legs and admissible recovery. 63-64: energy metrics, a 30-50+ series or justified N, accounting for repeats and dependence. 65: readiness of scenarios, instruments and criteria. These 5 analysis hours are split into 2 + 2 + 1.

66-68: task 9 and control faults first in HIL, then necessary rig checks. 69-71: the prepared series of short trials, including attempts already accepted; resetting conditions and cooling are included. 72-73: task 10 energy measurements and a service operation on the de-energised machine. 74-75: results table, success-rate interval, limitations and the six-legged robot v1.0 package, or a candidate with release blockers. Do not shorten checks merely to reach a repeat count: revise claimed precision and coverage.

Acceptance, additional gaits and the M24 handoff

Required evidence for v1.0

The package includes the six-legged-frame revision and 18-joint map; mass and centre of mass; power and communication calculations and measurements; repeated independent-stop verification; confirmed 120-second standing; wave and ripple with admissible phases; levelling within a defined region; responses to camera loss and node failure; a test series with justified N; an energy report; and inspection and maintenance instructions.

Every result has a raw file, configuration version and criterion. Holding level becomes a measured roll/pitch error, surface condition and duration. Traversing uneven terrain requires measured obstacle geometry and attempt count. Operation without a camera applies only to the tested restricted mode. An unmeasured quantity remains unknown even when the video appears convincing.

When to investigate tripod

Consider tripod only after verifying wave and ripple, actuator loading and feasibility of three-legged support. Typical alternating groups for a symmetric six-legged body are LF/RM/LR and RF/LM/RR; verify this convention against your names and geometry. For every phase, check contacts, margin and transition conditions. A triangle's existence does not guarantee admissible torques, friction or stability during swing.

Optional task if time permits: compare tripod and ripple phase tables in simulation with the same input, saving minimum margin and errors. A physical run requires a separate positive decision for that mode; it is outside the required M23 evidence. Similarly, five-legged walking remains research unless its operating envelope is calculated and verified.

Defence and handoff package

Answer: why do 18 actuators change the communication deadline; why is six-legged load not always shared equally; what does insufficient rank in plane estimation mean; how do loss of contact and slip differ; why does zero tilt not guarantee an admissible pose; what response is available for a jammed leg; what does E/d measure; and why do 40 successful attempts not mean 100% reliability? Demonstrate the answers using your data.

Pass to M24 requirements_candidates.csv, the attempt register, links to every raw log, assembly and calibration instructions, limitations, known defects and repair priorities. Record the hand's status and OPT results on a separate line. Do not rename an unfinished function as verified to meet the shared date. G4 remains a separate 35-hour integration checkpoint after the modules; it is not hidden within these 75 hours.

Assigned reading: six legs, contact and reliability

The references may be in English; the exercises and definitions are presented here in English. Read only the named sections before the associated work, within the existing 75 hours.

Core reading: contact and loading

1. R. Tedrake - Underactuated Robotics, MIT notes. Planning and Control through Contact → (Autonomous) Hybrid Systems. For tasks 3-4, write out foot modes, transition conditions (guards) and state changes on switching (reset); a phase transition requires a verified event. Retain the remaining contact-optimisation material as reference.

2. K. Lynch, F. Park - Modern Robotics. 12.1.2 Contact Types; 12.1.3 Multiple Contacts; 12.2.1 Friction. For tasks 4-6, distinguish touching, slipping and contact loss. Apply the contact model to feet; manipulation results do not directly establish six-legged stability.

A decision-making reference

3. NASA - Systems Engineering Handbook. 6.4 Technical Risk Management; 6.5 Configuration Management. For task 9 and PASS/HOLD, connect the fault event, consequence, test and assembly revision. Use a brief register without unsupported probabilities.

Documentation as needed

Actuator bus. DYNAMIXEL Protocol 2.0: Sync Read/Write, Bulk Read - only if it matches the real protocol. For task 2, recalculate communication volume and time for 18 devices; spares are excluded from active polling.

ROS diagnostics. README: diagnostic_updater, diagnostic_aggregator. For task 9, distinguish a missing reply from one leg, common-channel failure and stale status; check the branch against Jazzy.

NumPy. linalg.lstsq: rank, singular values, residuals. For task 5, assess geometric degeneracy in plane estimation. An empty residuals array does not establish a perfect fit.

Optional: architecture comparison

Kevin Ochs, ROS Hexapod Stack: Nodes and hexapod_controller. This historical ROS project is useful for comparing ripple and tripod; do not install its dependencies and gains into the current working environment. Reading deliverable: two transferable design choices and two incompatible assumptions.

Terms and definitions

Gait phase. Normalised position within a repeating cycle; it specifies the contact plan but does not confirm actual foot-surface contact.

Stance. An interval of planned leg contact during which admissible support force is constrained by traction, geometry and actuators.

Swing. Movement of a foot to its next support without planned support-force transmission; premature contact requires a separate response.

Wave gait. A slow sequential gait with legs swinging in turn; the specific order and admissible overlaps are defined by a phase table.

Ripple gait. A gait with phase-shifted waves of swing; the selected table, rather than the name, determines the number of simultaneous swings.

Tripod gait. A gait alternating two groups of three legs; group geometry alone does not guarantee admissible forces.

Guard. A testable condition for changing a discrete mode, such as confirmed contact or a timeout; activation is associated with a defined response.

Reset. A specified model or controller state change at a mode transition, such as clearing a timer and initialising a new support point.

Support polygon. The convex hull of selected contact points in a planar model; its applicability depends on assumptions about surface and forces.

Static stability margin. The minimum signed distance from the centre-of-mass projection to the support-polygon boundary, positive inside. Its use is limited to a planar quasi-static model.

Duty factor β . The fraction of a cycle a leg spends in planned stance; a dimensionless quantity between zero and one.

Surface normal. A unit vector perpendicular to a local plane; fix its direction and coordinate frame to prevent sign reversal of the estimate.

System rank. The number of independent data or constraint directions; insufficient rank means the selected plane parameters cannot be determined uniquely.

Slip. Relative tangential motion between a contacting foot and surface; a contact sensor may remain active during slipping.

Active levelling. Changing leg commands to reduce body-orientation error while respecting reachability, contacts and load limits.

Load redistribution. Changing forces among stance legs during movement or failure; additional contacts do not automatically equalise forces.

Bus utilisation. The fraction of available channel time occupied by communication, including overhead, pauses and retries; distinct from useful data throughput.

Specific energy E/d . Energy used per unit distance travelled, usually J/m; comparison requires identical conditions and a non-zero measured distance.

Relationships: $\beta = T_{\text{stance}}/T_{\text{cycle}}$; $E = \int U(t)I(t)dt$. Check: verified 120-second standing characterises the completed trial, not every possible duration.

Six-legged robot purchases and setup

USE EXISTING EQUIPMENT

Use the accepted M22 quadruped with 12 working actuators and 2 separate spares, onboard computer, MCU, IMU, camera, bus, instruments and support rig. For the existing 3D printer, choose the exact-model profile; print new parts to the six-legged-frame revision.

BUY BEFORE EXPANSION - AFTER M22 PASS

6 additional working actuators from the same verified family and **2 additional spares**. This gives **18 installed + 4 separate spares** if the earlier spares are unused. Update inventory when replenishing used spares. The M7 actuator rig is excluded from this count.

1 set of parts for two legs and a new or reinforced frame, bearings, inserts, fasteners and harnesses, with quantities from released CAD and BOM. If retaining M22 contact-sensing feet, add **2 channels for two feet**: $4 + 2 = 6$ total. Without the original four, select a separate six-channel set after testing one foot.

OPTIONAL OR SUBJECT TO CALCULATION

Recalculate power, thermal behaviour and communication for the new mass and 18 devices. Replace battery, BMS, charger, power disconnection, DC-DC converter, cables and protective devices only when they fail the measured budget; upgrading the battery alone is insufficient. Ratings depend on peaks and their durations, voltage drop and interruption currents, not the ratio 18/12.

Check the existing rig's load capacity and onboard-computer cooling. A second computer, CUDA GPU and new actuator family are optional. Hardware tripod operation requires a separate check of that mode after wave and ripple; do not expand while on HOLD.

INSTALL NOW

No new software stack is needed: pinned Ubuntu 24.04, ROS 2 Jazzy, `ros2_control`, NumPy and SciPy. If diagnostics are not connected, add `diagnostic_updater` and `diagnostic_aggregator` from [ROS diagnostics](#) for your version; check dependencies with `rosdep`. Retain the analysis and replay environment on the Mac and in UTM.

INSTALLATION AND CONFIGURATION CHECK

Before applying torque: check 18 unique IDs, 18 correct URDF joints, the software simulator, replay with a missing node and displayed diagnostics. Then measure communication cycle and stopping on the support. For terrain, check degenerate point sets, [REP-103 units and axes](#) and [SciPy proportion_ci: method, confidence_level](#) using the log of all attempts.

Preparation and checks fit within the existing 75 hours, replacing repeated reading and setup. Delivery waiting time does not count as study time.

Motion fundamentals: support transitions and equilibrium

Required analysis: support changes, body acceleration and actual contact

Walking changes both leg angles and the set of active contacts. A lifted foot no longer supplies support force; count a new support only after confirmed touchdown. The gait timer specifies intent; the sensor confirms the physical transition. Even six functioning legs do not guarantee stability when these transitions occur in the wrong order.

From static projection to a moving body

For slow standing on one horizontal plane, check that the centre-of-mass projection lies inside the contact polygon, friction is sufficient and loads are admissible. Under acceleration, this is insufficient. In a simplified planar model with constant height h and zero rate of change of angular momentum about the centre of mass: $x_{\text{CoP}} = x_c - (h/g) \cdot a_x$. CoP is the point of application of the resultant normal reaction; it must lie in the admissible support region.

Numerical counterexample. $x_c = 0.03$ m, $h = 0.25$ m, $g = 9.81$ m/s²; after a support transition, the admissible longitudinal CoP interval is $[-0.08; 0.12]$ m. At rest, $\text{CoP} = 0.03$ m. Requesting $a_x = 5$ m/s² gives $\text{CoP} = -0.09742$ m: the centre-of-mass projection is inside, but the required reaction lies outside. This model rejects the command. This is a calculation test, not an instruction to accelerate the physical robot.

For $\mu = 0.6$, the aggregate friction constraint gives $|a_x| \leq \mu g = 5.886$ m/s² when vertical acceleration is zero. It separately permits 5 m/s², but the CoP condition is already violated. Both conditions and actuator constraints must hold. A longitudinal interval does not check lateral stability; use complete contact geometry for the robot.

Practice: verify a wave → ripple transition

In the existing simulation scenario, select the instant before leg lift-off. Construct current and next support regions using confirmed contacts only and calculate static margin. Then estimate required CoP under the declared assumptions; for the general case, use M20 force constraints. Inject loss of one contact: defer swing or select a previously verified fallback.

Save `contact_transition.csv` and support diagrams: phase commands, touchdown confirmations, x_c , CoP estimate, boundaries and decision reason. Separately test delayed swing and a false early touchdown. Increasing duty factor provides more time in stance, but does not establish that every transition is feasible or friction sufficient.

Within 75 hours: 1 hour from 23-24 and 2 hours from the simulation portion of 31-34, tasks 3-4. Physical trials remain within the accepted M22 region. Read Russ Tedrake, [Highly-articulated Legged Robots: CoP and ZMP](#) - assumptions of simplified dynamics. Do not apply this formula to contacts at different heights or large body rotations without checking its validity.